MECHANICAL ENGINEERING PORTFOLIO | 2026

HASHIM KHAN

Mechanical Design Engineer

Robotics · Automation · Electromechanical Systems · Manufacturing

I’m a Senior Mechanical Design Engineer with 9 years of experience across robotics, industrial automation, defence, marine systems, and electromechanical product development. I specialise in mechanical system design, mechanism development, CAD, engineering analysis, and prototype validation.

My multidisciplinary background enables me to approach engineering challenges with a broad perspective, balancing performance, reliability, and manufacturability. I enjoy collaborating with international, cross-functional teams to transform innovative concepts into practical, production-ready solutions. I’m passionate about robotics, emerging technologies, and continuous engineering improvement.

RVM & Compactor









Project Overview

Mechanical design and development of reverse vending machines, compaction mechanisms, and conveyor systems for automated bottle and can handling. The work involved developing mechanical assemblies, evaluating motion mechanisms, integrating drive components, and supporting prototype testing and manufacturing optimisation.

Key Engineering Contributions

1. Mechanical Architecture & CAD Design

- Developed mechanical assemblies, structural frames, sheet-metal enclosures, brackets, and component interfaces.
- Translated functional requirements into CAD models, detailed drawings, and manufacturable designs.
- Considered packaging constraints, assembly access, maintainability, and integration with sensors, cameras, and control systems.

2. Compactor Mechanism & Force Analysis

- Evaluated compaction mechanism geometry, roller positioning, load paths, and motion requirements.
- Considered the influence of bottle geometry, material behaviour, contact conditions, and operating loads on compaction performance.
- Assessed critical mechanism positions and potential jam conditions to support reliable operation.

3. Torque Calculation & Motor/Gearbox Selection

- Evaluated drive requirements based on operating loads, rotational speed, transmission ratio, and mechanism geometry.
- Considered torque requirements, gearbox efficiency, starting loads, and duty conditions when selecting drive components.
- Integrated a hypoid gearbox into the compaction drive system, considering mounting, packaging, and mechanical interfaces.

4. Material Selection & Structural Design

- Considered strength, stiffness, wear resistance, corrosion, mass, cost, and manufacturing feasibility during material selection.
- Designed structural frames, sheet-metal components, brackets, and mechanical interfaces to support operating loads.
- Applied engineering calculations, load-path assessment, GD&T, and tolerance stack-up principles.

5. FEA & Design Verification

- Engineering analysis considerations included structural loading, stress distribution, displacement, and component behaviour.

- Relevant verification activities include defining material properties, boundary conditions, load cases, and acceptance criteria.
- Used engineering analysis and prototype feedback to guide design improvements where applicable.

6. Adjustable V-Type Conveyor Design

- Developed a dual-belt V-type conveyor concept for controlled bottle guidance and material handling.
- Considered V-angle adjustment, belt tracking, tensioning, roller alignment, bottle stability, and frame stiffness.
- Considered adjustment locking, service access, and integration with upstream inspection and downstream handling systems.

7. Electrical Integration & EMI/EMC Considerations

- Coordinated mechanical interfaces for sensors, cameras, motors, cable routing, and control-system integration.
- Considered enclosure bonding, cable segregation, grounding interfaces, and separation of power and sensitive signal wiring as part of system integration.
- EMC compliance testing and shielding effectiveness should be reported separately if performed and documented.

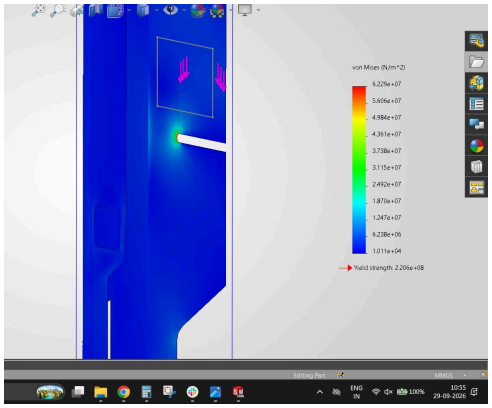
Prototype Testing & Design Improvement

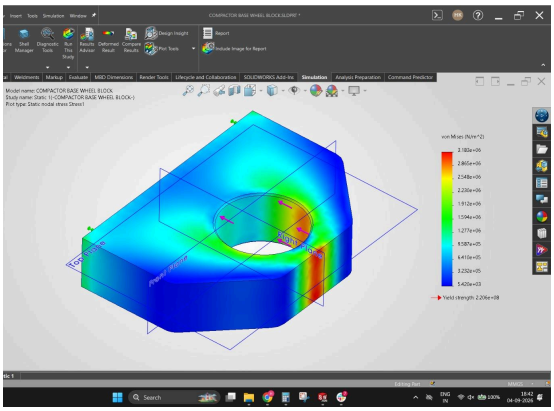
During functional testing of an RVM compactor, bottle jamming was identified and eliminated by repositioning the rollers 20° upward. Retesting confirmed the improvement. The compactor assembly mass was reduced from 110 kg to 80 kg, representing a **27.3% reduction**.

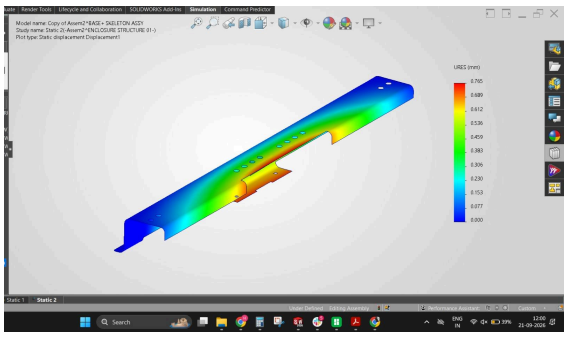
Production output also increased from 12 to 18 machines per day, a **50% improvement**, through engineering and production improvements.

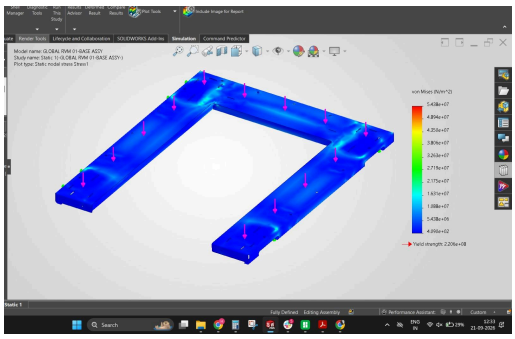
Engineering Value

- Mechanism design and mechanical system integration
- Drive-system evaluation and gearbox integration
- Design optimisation for mass, reliability, and manufacturability
- Prototype testing, troubleshooting, and root-cause analysis
- Cross-functional collaboration with manufacturing and electrical/control-system teams









AI Vision Based counting & Inspection systems





1. System Architecture & CAD Design

- Took end-to-end ownership of mechanical system design, from concept development and 3D CAD modelling to detailed design and component integration.
- Developed a three-level, multi-conveyor architecture integrating the hopper, conveyor assemblies, inspection stations, and supporting structures within a compact footprint.
- Created detailed CAD assemblies and component interfaces, reviewing clearances, assembly feasibility, and potential mechanical interference before fabrication.

2. Conveyor Design & Drive Selection

- Designed three conveyor sections of **0.5 m, 1.5 m, and 0.8 m**, using PVC belts to support material transfer through the inspection process.
- Integrated **0.5 HP motors with 1:20 worm gearboxes**, considering conveyor drive requirements and the intended material-handling application.
- Performed conveyor-related engineering calculations and considered belt tracking, tensioning, roller alignment, frame stiffness, and transfer-point geometry.

3. Hopper & Material Handling Design

- Designed the hopper and associated mechanical interfaces to guide incoming material towards the conveyor system.
- Developed the material-transfer layout between the hopper, conveyor sections, and inspection zones to support a continuous material-handling sequence.
- Considered hopper geometry, component clearances, equipment packaging, and accessibility for cleaning and maintenance.

4. Machine Vision & Sensor Integration

- Designed mechanical mounting arrangements for cameras, illumination units, and sensors used in AI-based counting and inspection.
- Developed mounting brackets and interfaces to support suitable positioning, alignment, and integration of vision-system hardware with the conveyor layout.
- Considered lighting and camera access, sensor placement, cable routing, and mechanical clearances in coordination with electronic system requirements.

5. Structural Design & Manufacturing

- Designed supporting frames, sheet-metal components, mounting brackets, and mechanical interfaces for the conveyor and inspection assemblies.
- Developed detailed CAD models for sheet-metal fabrication and CNC-machined components, considering manufacturing processes and component geometry.
- Incorporated structural rigidity, fastening arrangements, assembly access, and component alignment into the mechanical design.

6. DFM/DFA & Engineering Validation

- Applied Design for Manufacturing (DFM) and Design for Assembly (DFA) principles to improve fabrication feasibility, assembly simplicity, and subsystem integration.
- Reviewed conveyor alignment, belt tensioning, roller positioning, fastening interfaces, and assembly clearances during design development.
- Considered serviceability, maintenance access, and integration with camera, lighting, sensor, and cable-routing requirements.

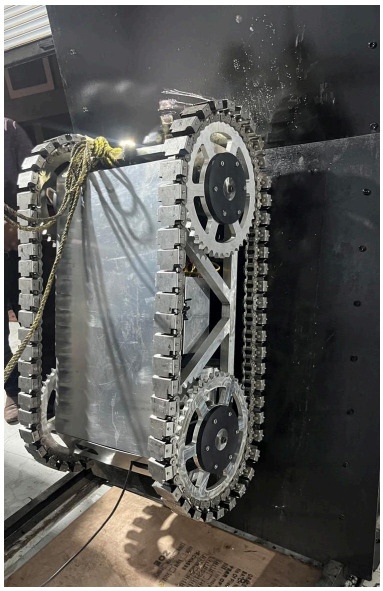
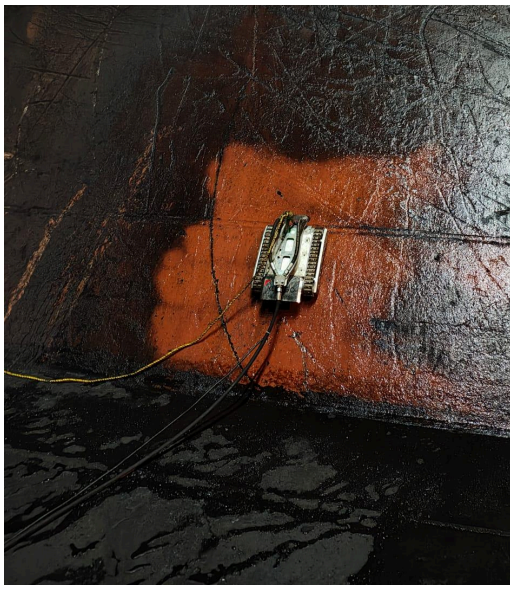
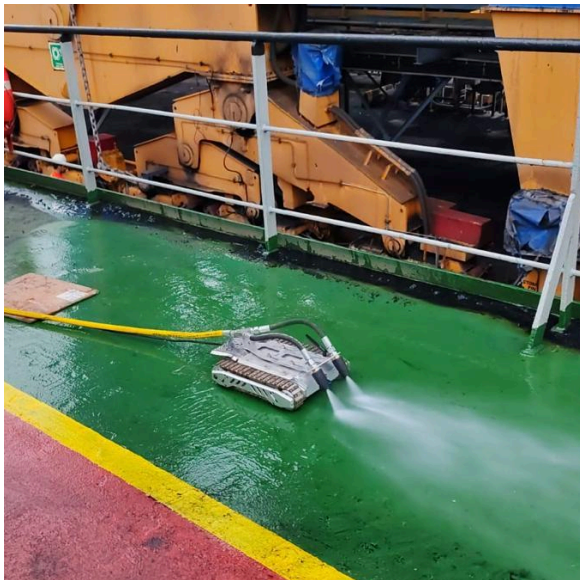
Key Design Specifications

Parameter	Specification
Application	AI vision-based counting and inspection
Conveyor configuration	Three-level, multi-conveyor system
Conveyor lengths	0.5 m, 1.5 m, and 0.8 m
Combined conveyor length	2.8 m
Belt material	PVC
Motor rating	0.5 HP
Gearbox type	Worm gearbox
Reduction ratio	1:20
Manufacturing processes	Sheet-metal fabrication and CNC machining

Engineering Contribution

Owned the mechanical design from system architecture and conveyor development to machine vision hardware integration and manufacturing-oriented detailing. Coordinated mechanical interfaces with vision-system hardware, electronics, and manufacturing requirements to develop an integrated inspection machine.

ROV - Cargo Ship Hull Cleaning and Inspections

1. System Architecture & CAD Design

- Led mechanical R&D and product development of a remotely operated magnetic crawler ROV for marine applications, including cargo-ship hull cleaning, inspection, and maintenance.
- Developed the mechanical architecture around a compact crawler chassis, magnetic adhesion system, track-and-sprocket arrangement, drive mechanisms, and cleaning-tool interfaces.
- Designed mechanical assemblies for operation in seawater environments, considering packaging constraints, structural integrity, maintainability, and integration of electrical and control-system hardware.

2. Drive System Design & Gearbox Selection

- Evaluated drive-train configurations using motor torque, speed, load, duty cycle, and packaging requirements to establish suitable propulsion and crawler movement characteristics.
- Compared **planetary, cycloidal, and strain-wave (harmonic) gearboxes**, considering torque density, reduction ratio, backlash, efficiency, durability, and installation space.
- Worked on drive-system sizing and the selection of motors, bearings, sprockets, and transmission components to support controlled movement and manoeuvrability on ship surfaces.

3. Magnetic Adhesion & Cleaning Mechanism

- Developed a magnetic crawler concept using dual magnetic tracks to maintain adhesion to ferromagnetic ship surfaces during hull-cleaning operations.
- Evaluated neodymium permanent-magnet arrangements, magnet contact distribution, crawler loading, and detachment conditions to support adhesion-system design.
- Considered integration of cleaning brushes or high-pressure jet-based cleaning tools, with mechanical interfaces designed around the crawler architecture and intended cleaning application.

4. Mechanical Design & Engineering Calculations

- Designed and evaluated chain-and-sprocket, gear-and-pinion, and screw-drive mechanisms, including component selection and mechanical interface development.
- Performed engineering calculations covering torque, power, kinematics, dynamics, bearing selection, and drive requirements to assess the feasibility of the mechanical system.
- Considered chassis materials, corrosion resistance, sealing arrangements, structural loading, and component accessibility for reliable operation in demanding marine environments.

5. Structural Design & Manufacturing

- Evaluated materials and construction approaches, including stainless steel 316L and anodised aluminium, for structural components and marine-exposed assemblies.
- Developed CAD models and mechanical assemblies with attention to fabrication, machining, fastening, component integration, and assembly access.
- Considered sealed enclosures, pressure-resistant housing concepts, and double O-ring sealing arrangements to protect sensitive hardware against seawater ingress.

6. DFM/DFA & Engineering Validation

- Applied Design for Manufacturing and Assembly principles to improve component integration, assembly feasibility, maintainability, and overall mechanical packaging.
- Used engineering analysis, including FEA and fluid-system assessment where applicable, to support design evaluation and refinement of the mechanical architecture.
- Considered cross-functional integration of mechanical systems with motors, sensors, encoders, electronics, and control hardware during product development.

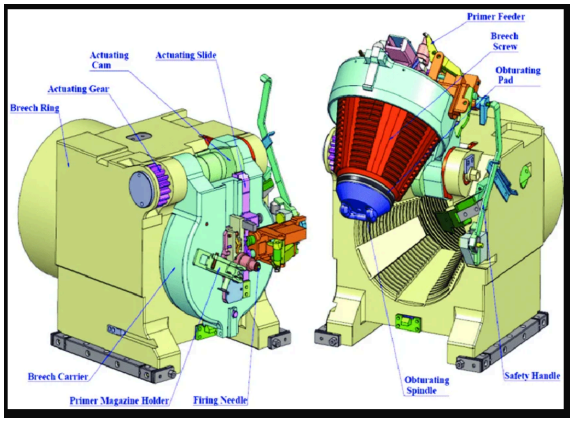
Key Design Specifications

Parameter	Design details
Application	Cargo-ship hull and cargo-hold cleaning
Platform	Remotely operated magnetic crawler ROV
Adhesion concept	Dual magnetic tracks
Magnet type	Neodymium N52 permanent magnets
Indicative design mass	100–120 kg
Marine environment	Seawater operation
Gearbox technologies evaluated	Planetary, cycloidal, strain-wave
Mechanical systems	Chassis, crawler drive, sprockets, transmission, cleaning interfaces
Engineering scope	Mechanical R&D, CAD, calculations, component selection, design validation

Engineering Contribution

Led mechanical R&D activities spanning crawler architecture, magnetic adhesion, transmission design, gearbox evaluation, engineering calculations, and marine-oriented mechanical design. The project combined robotics, precision mechanisms, electromechanical integration, and manufacturability considerations to address the challenges of remotely operated ship-cleaning equipment.

Artillery Systems - Barrel, Breech,Tooling, Recoil systems











1. System Architecture & CAD Design

- Contributed to mechanical engineering and manufacturing activities for defence artillery programmes, including ATAGS 155×52, MGS 155×39, Garuda 105, and SRGM 76.
- Worked on mechanical assemblies associated with barrel ordnance, shell-loading mechanisms, and breech mechanisms, supporting component design, integration, and manufacturing requirements.
- Applied engineering drawing practices, dimensional control, GD&T, and interface management to support design interpretation, component compatibility, and production readiness.

2. Mechanism Design & Engineering Calculations

- Supported mechanical design and engineering assessment of artillery subassemblies, focusing on functional interfaces, component fit, manufacturability, and mechanical reliability.
- Considered loading mechanisms, moving interfaces, clearances, tolerance requirements, and assembly constraints during design and manufacturing reviews.
- Coordinated design requirements with manufacturing and quality teams to address component-level issues and improve production feasibility.

3. Material Selection, Hardness & Surface Treatment

- Worked with material specifications and component requirements for defence-grade mechanical assemblies, considering strength, wear resistance, hardness, and service conditions.
- Supported the application and verification of specified heat-treatment, hardness, and surface-treatment requirements in accordance with approved engineering documentation.

- Considered material traceability, inspection requirements, dimensional stability, and surface-finish requirements as part of manufacturing and quality coordination.

4. Barrel Ordnance & CNC Manufacturing

- Gained experience in manufacturing engineering activities associated with barrel ordnance and precision-machined defence components, with emphasis on dimensional accuracy and process control.
- Supported CNC machining planning and tooling development, considering workholding, tool accessibility, machining sequence, setup requirements, and component inspection.
- Coordinated with manufacturing teams to address machining challenges, drawing requirements, process feasibility, and quality expectations for complex components.

5. Standards, Quality & Manufacturing Documentation

- Worked within controlled defence engineering and manufacturing environments requiring adherence to applicable MIL-STD, DEF-STAN, ISO, and programme-specific requirements.
- Considered GD&T, dimensional inspection, material certification, hardness verification, surface-finish requirements, and manufacturing documentation as part of quality assurance.
- Coordinated with design, production, and quality teams to support technical reviews, non-conformance resolution, drawing compliance, and production documentation.

6. Tooling Development & Production Planning

- Supported tooling and fixture development for CNC machining and component manufacturing, considering repeatability, setup consistency, accessibility, and process reliability.
- Contributed to production planning through manufacturing-sequence reviews, process coordination, tooling readiness, and alignment of engineering requirements with shop-floor capabilities.
- Collaborated with cross-functional teams, including manufacturing engineers, CNC programmers, quality inspectors, and production personnel, to address technical issues and support production readiness.

Engineering Contribution

Contributed to defence engineering programmes spanning mechanical assemblies, barrel ordnance, loading and breech mechanisms, precision CNC manufacturing, tooling development, and production planning. The experience strengthened my understanding of controlled engineering documentation, precision manufacturing, material and surface-treatment requirements, quality assurance, and cross-functional collaboration in a highly regulated environment.

Inspection Systems - Crankshaft Inspection using MPI-Ai Imaging



1. System Architecture & CAD Design

- Designed mechanical subsystems for a crankshaft inspection machine integrating Magnetic Particle Inspection (MPI), AI-based imaging, and automated material handling.
- Developed CAD assemblies for camera motion systems, crankshaft clamping mechanisms, rotary drives, and robotic pick-and-place grippers.
- Prepared mechanical layouts, mounting arrangements, and component interfaces to support system integration and manufacturing readiness.

2. Mechanism Design & Engineering Calculations

- Designed LM guide rail arrangements for camera positioning, linear drive systems for crankshaft clamping, and rotary mechanisms for controlled component rotation.
- Developed dual-clamp robotic grippers for automated pick-and-place operations, considering component handling, alignment, and mechanical interfaces.
- Performed engineering calculations and assessed component fit, clearances, motion requirements, and assembly constraints during design development.

3. Sensor Integration, Lighting & Cable Management

- Collaborated with IIoT engineers to integrate sensors and mounting provisions for Industry 4.0-enabled inspection systems.
- Designed mounting arrangements for white and UV-A LED lighting, including UV radiation shields to support the inspection setup.
- Developed cable-routing provisions and mechanical-electronics interfaces to improve accessibility, integration, and maintainability.

4. Mechanical Development, Material Selection & Validation

- Managed end-to-end mechanical component design, including material selection, component development, engineering calculations, and assembly integration.
- Coordinated with cross-functional teams to address mechanical and electronics integration requirements throughout machine development.
- Supported Factory Acceptance Testing (FAT) and Site Acceptance Testing (SAT) in accordance with applicable ISO standards and project requirements.

